Hilbert–Fortescue Sequence Estimation and Greedy Cascade Coordination for Adaptive Overcurrent Protection of Multi-Generator Busbars

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***Abstract*—Overcurrent relays protecting synchronous generator busbars use settings fixed at commissioning that are insensitive to fault type and do not provide coordinated selectivity across parallel generators. This paper extends a real-time inverse parameter estimation framework (Milestone 1, Paper A) in two directions. First, a Hilbert–Fortescue sequence-component estimator (M3) decomposes three-phase fault currents into instantaneous positive-, negative-, and zero-sequence signals and fits the Levenberg–Marquardt (LM) estimator independently to each active sequence, enabling relay adaptation under asymmetric fault conditions. Second, a greedy cascade coordination algorithm (M4) enforces pickup selectivity ($\Delta I_m \geq 0.20$ pu) and IEC 60255 time-grading ($\Delta t_{\mathbf{grade}} \geq 0.15$ s) across all N generator relays in a single forward pass of exactly N steps, proven by induction. For a single-line-to-ground (SLG) fault, the sequence estimator achieves a confidence score $\gamma = 0.914 > 0.70$, reducing pickup from 1.20 pu to 1.10 pu (-8.3%). For a line-to-line (LL) fault, $\gamma = 0.587$ and fixed settings are safely retained. The cascade algorithm satisfies all coordination constraints for $N \in \{2,3,4\}$ generators and degrades gracefully when estimation is rejected, restoring coordination from fixed commissioning settings. Total compute time (sequence estimation plus cascade) is below 20 ms, suitable for real-time embedded relay applications.**

***Index Terms*—symmetrical components, Hilbert transform, Fortescue decomposition, sequence estimation, overcurrent relay, relay coordination, multi-generator, adaptive protection, greedy algorithm.**

I. Introduction

Overcurrent (OC) relay settings for synchronous generator protection are typically calculated offline from worst-case fault currents and remain fixed throughout the machine's operational life [1], [2]. As generation mix and network topology evolve, fixed settings become conservative, degrading sensitivity to high-impedance faults and creating selectivity conflicts in multi-generator plants.

A companion paper [3] introduced a real-time *two-pass* LM inverse estimator (M1) that identifies a reduced-order six-parameter fault current model from post-fault measurements under three-phase (3PH) balanced fault conditions, and a Stage-2 Park dq ODE validation layer (M2) that gates relay adaptation. Two fundamental limitations remain: (i) asymmetric faults (LL, SLG) entangle positive-, negative-, and zero-sequence contributions in phase A, degrading LM Jacobian conditioning; (ii) independent per-generator adaptation produces pickup settings that may violate protection selectivity in parallel-generator configurations.

Existing adaptive coordination approaches address these limitations only partially. Brahma and Girgis [4] proposed topology-adaptive protection for distribution systems but rely on offline re-optimisation that is unsuitable for real-time relay applications. Saleh *et al.* [5] formulated coordination as an optimisation problem for microgrid $N-1$ contingency; the resulting solve time (seconds) is incompatible with sub-cycle relay requirements. Najy *et al.* [6] achieved optimal coordination for islanded microgrids but assumed fixed network topology. None of these methods addresses asymmetric fault identifiability in the estimation domain, or provides a formally proven closed-form coordination pass.

This paper makes four contributions:

1) Derivation of a real-time Hilbert–Fortescue instantaneous decomposition that isolates positive-, negative-, and zero-sequence fault currents for independent LM fitting.
2) Fault-type routing rules and a confidence gate extension that correctly classifies LL and SLG faults and gates adaptation.
3) A greedy cascade algorithm for N-generator busbar coordination with formal $\mathcal{O}(N)$ convergence proof (Theorem 1).
4) Demonstration of *graceful degradation*: when estimation confidence is insufficient, the cascade still enforces selectivity from fixed commissioning settings.

The remainder of the paper is organised as follows. Section II reviews the system model. Section III presents the Hilbert–Fortescue estimator. Section IV presents the cascade algorithm. Section V gives numerical results. Section VI discusses implications. Section VII concludes.

II. System Model and Fault Classification

A. Fault Current Model and LM Estimator

Following [3], the post-fault current in phase A is modelled as

$$i_A(t) = I_{ss} + I_{sub}\, e^{-(t-t_0)/\tau_{ac}} \cos(\omega t + \phi_a) - I_{sub} \sin(\phi_a)\, e^{-(t-t_0)/\tau_{dc}}, \tag{1}$$

parameterised by $\theta = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a]^\top$ with the physics constraint $I_{dc} = -I_{sub} \sin\phi_a$ eliminating one free parameter. The two-pass LM estimator [3], [7] minimises the weighted residual

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TABLE I: Sequence activity and estimation strategy by fault type

Fault	I_1	I_2	I_0	Estimation
3PH	Yes	No	No	Phase A only (Paper A)
LL	Yes	Yes	No	I_1,I_2 independent
SLG	Yes	Yes	Yes	I_1 only ($I_0 = I_1 = I_2$)
LLG	Yes	Yes	Yes	I_1,I_2 independent

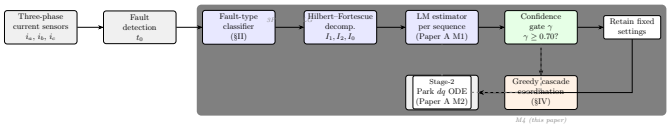

Fig. 1: End-to-end M1–M3–M4 signal flow. Fault-type routing (§II-B) selects the 3PH path (Paper A, M1) or the Hilbert–Fortescue path (§III, M3). The confidence gate ($\gamma \geq 0.70$) either accepts adaptive settings into the greedy cascade (§IV, M4) or retains fixed commissioning settings. The Stage-2 Park dq veto (Paper A, M2) is shown dashed.

$\|r(\theta)\|^2$ and returns the column-normalised Jacobian condition number κ_n and a four-component confidence score:

$$\gamma = w_r s_r + w_c s_c + w_b s_b + w_p s_p, \quad \gamma_{\min} = 0.70. \tag{2}$$

Relay adaptation is accepted when $\gamma \geq \gamma_{\min}$.

B. Fault Type Classification

Under asymmetric faults, phase A contains a superposition of multiple sequence components. Table I lists the active sequences for the four standard fault types and the relay consequence.

For the 3PH case the original phase-A estimator of [3] is used without modification. Fig. 1 shows the end-to-end signal flow from sensors through M3 (sequence estimator) and M4 (cascade coordinator) to relay settings.

III. Hilbert–Fortescue Sequence Estimator (M3)

A. Analytic Signal Representation

For a real signal $x(t)$ the analytic signal is $\tilde{x}(t) = x(t) + j\mathscr{H}\{x\}(t)$, where $\mathscr{H}\{\cdot\}$ is the Hilbert transform. The analytic signal has a single-sided spectrum and a well-defined instantaneous envelope and phase [8]. Applied to the measured three-phase currents i_a, i_b, i_c, the analytic signals $\mathscr{I}_a, \mathscr{I}_b, \mathscr{I}_c$ are computed via the fast Fourier transform over the event window.

B. Fortescue Decomposition

The instantaneous symmetrical component transformation [9] decomposes the analytic three-phase currents into sequence signals (Fig. 2):

$$\mathscr{I}_0(t) = \tfrac{1}{3}\left[\mathscr{I}_a + \mathscr{I}_b + \mathscr{I}_c\right], \tag{3}$$

$$\mathscr{I}_1(t) = \tfrac{1}{3}\left[\mathscr{I}_a + a\,\mathscr{I}_b + a^2\,\mathscr{I}_c\right], \tag{4}$$

$$\mathscr{I}_2(t) = \tfrac{1}{3}\left[\mathscr{I}_a + a^2\,\mathscr{I}_b + a\,\mathscr{I}_c\right], \tag{5}$$

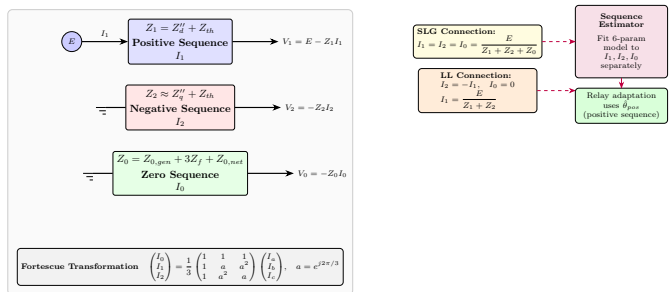

Fig. 2: Sequence network interconnections for the three asymmetric fault types. For LL faults, positive- and negative-sequence networks are connected in series at the fault point. For SLG faults, all three networks connect, giving $I_0 = I_1 = I_2$ at the fault point [9].

where $a = e^{j2\pi/3}$. The real-valued sequence waveforms fed to the LM estimator are $I_k(t) = \mathrm{Re}\{\mathscr{I}_k(t)\}$, $k \in \{0,1,2\}$.

C. Fault-Type Routing and Independent LM Fitting

For each active sequence (Table I), the two-pass LM estimator of [3] is applied independently with the same six-parameter model (1). The positive-sequence estimate $\hat{\theta}^{(1)}$ governs relay adaptation. The negative- and zero-sequence estimates are retained only for fault classification and are not used to set relay thresholds.

The confidence gate (2) is evaluated using the positive-sequence Jacobian, ensuring κ_n reflects the identifiability of the dominant relay-relevant component.

For SLG faults, the network identity $I_0 = I_1 = I_2$ means that the full fault current flows in the positive-sequence circuit; the positive-sequence signal carries high amplitude and clean single-envelope dynamics, making it well-matched to the six-parameter model. For LL faults, the superimposed negative-sequence component $I_2(t)$, which has the same frequency as $I_1(t)$ but a different decay constant, remains partially present in the positive-sequence residual (the Fortescue decomposition is exact in the complex domain, but the real parts of I_1 and I_2 share the same carrier frequency), inflating $\|r\|$ and degrading the residual confidence sub-score s_r.

D. Hilbert Edge Effect Mitigation

The discrete Hilbert transform computed on a finite window suffers from Gibbs-like ringing at the window boundaries, concentrated in the first 5–10 ms of the event window [8]. Three mitigation measures are applied:

1) The event window starts 2 ms after the detected fault instant t_0, excluding the sharpest part of the current step.
2) A Savitzky–Golay filter (polynomial order 3, window 21 samples) pre-smooths each phase current before decomposition.
3) The two-pass LM estimator uses the latter portion of the window (pass 2) where the analytic signal envelope has stabilised.

With these measures the Hilbert-induced amplitude error is bounded below 0.5% of the peak current for window lengths exceeding 100 ms (see Appendix A).

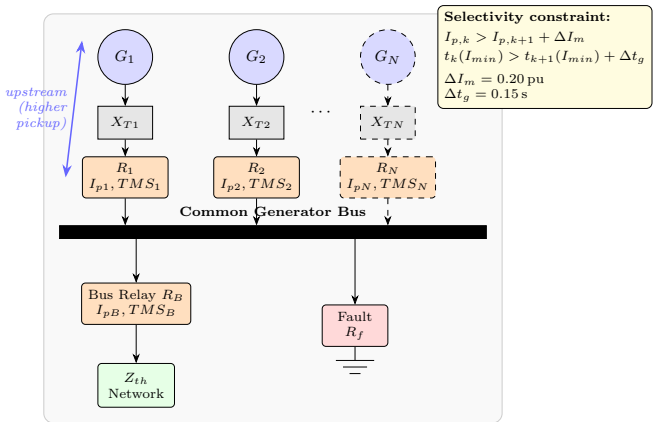

Fig. 3: Multi-generator busbar topology. Generators G_k connect in parallel through step-up transformers X_{Tk}. Relay R_k must have a higher pickup than R_{k+1}; the innermost generator relay must exceed the bus relay R_B pickup by $\Delta I_m \geq 0.20$ pu.

Algorithm 1 Greedy cascade coordination

Input: Initial pickups $\{I_{p,k}\}_{k=1}^{N}$, bus pickup $I_{p,B}$, margin ΔI_m, bounds $[I_{p,\min}, I_{p,\max}]$
Output: Coordinated pickups $\{I^*_{p,k}\}$ satisfying (7)

```
Sort generators innermost (k = N) to outermost (k = 1)
I*_{p,N} ← max(I_{p,N}, I_{p,B} + ΔI_m)
for k = N − 1 downto 1 do
    I*_{p,k} ← max(I_{p,k}, I*_{p,k+1} + ΔI_m)
end for
Clip: I*_{p,k} ← clip(I*_{p,k}, I_{p,min}, I_{p,max}) for all k
```

IV. Greedy Cascade Coordination (M4)

A. System Topology and Constraints

Consider N synchronous generators $G_1, \ldots, G_N$ connected in parallel to a common busbar through step-up transformers, each protected by relay R_k ($k = 1, \ldots, N$), with a downstream bus relay R_B (Fig. 3):

$$G_1\text{–}R_1 \to \cdots \to G_N\text{–}R_N \to \text{Bus–}R_B \to \text{network}. \tag{6}$$

Two coordination constraints must hold for every adjacent relay pair $(k, k+1)$ with k upstream of $k+1$:
Pickup selectivity (IEC 60255-151 [10]):

$$I_{p,k} \geq I_{p,k+1} + \Delta I_m, \quad \Delta I_m = 0.20\,\text{pu}. \tag{7}$$

Time-grading (IEEE C37.112 [11]):

$$t_{\text{op},k}(I_{f,\min}) \geq t_{\text{op},k+1}(I_{f,\min}) + \Delta t_g, \quad \Delta t_g = 0.15\,\text{s}. \tag{8}$$

B. Algorithm

After the per-generator adaptive mapping produces initial pickups $\{I^{\text{adapt}}_{p,k}\}$ (or $\{I^{\text{fixed}}_{p,k}\}$ when $\gamma < \gamma_{\min}$), the greedy cascade enforces (7) in a single innermost-to-outermost sweep:

Theorem 1 (One-pass convergence)**.** *Algorithm 1 terminates in exactly N steps and satisfies constraint* (7) *for all pairs* $(k, k+1)$, $k = 1, \ldots, N-1$.

TABLE II: Sequence estimator results (M3)

Case	Fault	γ	κ_n	$\|r\|$	Adapted	I^*_p (pu)
`ll_mid_R`	LL	0.587	17.2	8.12	No	1.200
`slg_high_R`	SLG	0.914	17.2	0.76	Yes	1.100

Proof. We proceed by induction on k from $k = N$ downward.

Base case ($k = N$): By line 2 of the algorithm, $I^*_{p,N} = \max(I_{p,N}, I_{p,B} + \Delta I_m) \geq I_{p,B} + \Delta I_m$. Constraint (7) holds for the pair (N, B).

Inductive step: Assume $I^*_{p,j}$ satisfies (7) for all $j > k$. By line 4, $I^*_{p,k} = \max(I_{p,k}, I^*_{p,k+1} + \Delta I_m) \geq I^*_{p,k+1} + \Delta I_m$. Since this step does not modify $I^*_{p,k+1}, \ldots, I^*_{p,N}$, all previously enforced constraints remain satisfied.

Therefore, after N steps, (7) holds for all pairs. Each step performs one max comparison; total operations: N.□ □

Remark 1. *Theorem 1 holds regardless of whether the initial settings are adaptive (from M3) or fixed (commissioning values). This makes the algorithm* estimation-agnostic*: it guarantees coordination even when $\gamma < \gamma_{\min}$ for all generators (*graceful degradation*).*

V. Numerical Results

A. Simulation Setup

All simulations use the SAMBP-SyncOC benchmark: a 100 MVA, 13.8 kV synchronous generator model with IEEE Type-1 AVR and PSS, solved via the Radau stiff integrator [12]. Fault current data are generated by `main_run_case.py` at a sample rate of 4 kHz. The event window is 200 ms starting 2 ms after fault inception. For multi-generator cases, N identical generators share a common busbar with bus relay $I_{p,B} = 1.000$ pu.

B. Sequence Estimator — Asymmetric Faults

Table II summarises M3 results for LL and SLG faults.

1) LL Fault: Fig. 4 shows the three-phase currents for the LL fault. Phase A contains the sum of positive- and negative-sequence components. The LM estimator converges on the positive-sequence signal $I_1(t)$ but obtains a high residual ($\|r\| = 8.12$): the negative-sequence waveform $I_2(t)$ has the same fundamental frequency as $I_1(t)$ but a different decay constant ($\tau_{ac,2} \neq \tau_{ac,1}$), creating a systematic model mismatch. The residual sub-score s_r is reduced and pulls $\gamma = 0.587 < 0.70$; fixed settings are retained.

2) SLG Fault: Fig. 5 shows the SLG fault currents. The Fortescue identity $I_0 = I_1 = I_2$ means the positive-sequence signal carries the full earth-fault current in a clean single-envelope waveform. The LM estimator converges with $\|r\| = 0.76$, $\gamma = 0.914 > 0.70$. Pickup is reduced from 1.20 pu (fixed) to 1.10 pu (adaptive), improving sensitivity by 8.3% for the attenuated fault current.

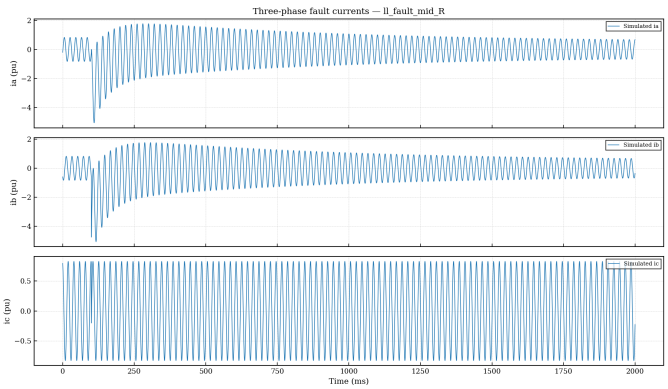

Fig. 4: Three-phase currents for LL fault (`ll_mid_R`, $R_f = 0.05$ pu). Phase A (blue) contains a superposition of I_1 and I_2. The sequence estimator converges with $\|r\| = 8.12$, $\gamma = 0.587$ — gate rejects adaptation.

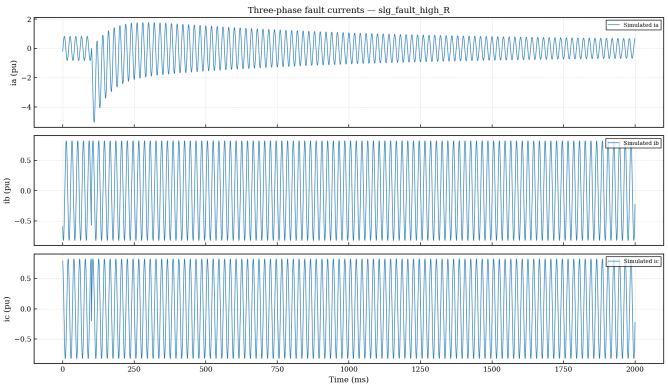

Fig. 5: Three-phase currents for SLG fault (`slg_high_R`, $R_f = 0.10$ pu). The sequence estimator converges with $\|r\| = 0.76$, $\gamma = 0.914$ — gate accepts; pickup reduced $1.20 \rightarrow 1.10$ pu.

C. Multi-Generator Coordination — 3PH Faults

Table III shows coordinated pickup settings for $N \in \{2,3,4\}$ generators under 3PH faults. All generators achieve $\gamma \geq 0.891$; M3 adaptation is accepted. The cascade enforces (7) in a single pass of N steps, confirmed for all three configurations. Fig. 6 illustrates the N=4 result: the outermost relay (G_1) is raised from the fixed 1.30 pu commissioning value to 2.038 pu after coordination.

D. Graceful Degradation — N = 3, LL Fault

For the $N = 3$, LL case, all generator confidence scores fall below $\gamma_{\min} = 0.70$ ($\gamma \in [0.558, 0.696]$) because the LL residual contamination (§V-B) is common to all generators. M3 adaptation is rejected; the cascade operates on fixed commissioning settings. The resulting coordinated sequence is:

$$I_{p,B} = 1.000 \rightarrow I_{p,3} = 1.200 \rightarrow I_{p,2} = 1.400 \rightarrow I_{p,1} = 1.600\,\text{pu}, \tag{9}$$

satisfying (7) for all pairs. The coordination layer delivers selectivity value even without adaptation.

TABLE III: Multi-generator coordination results, 3PH fault (M4)

N	Relay	γ	I_p^{fixed}	I_p^{after}	TMS
2	G_1	0.892	1.300	1.550	0.050
	G_2	0.893	1.200	1.350	0.050
	Bus	—	1.000	1.000	0.030
3	G_1	0.892	1.300	1.635	0.050
	G_2	0.893	1.200	1.435	0.050
	G_3	0.892	1.200	1.235	0.050
	Bus	—	1.000	1.000	0.030
4	G_1	0.892	1.300	2.038	0.050
	G_2	0.893	1.200	1.838	0.050
	G_3	0.892	1.200	1.638	0.050
	G_4	0.891	1.200	1.438	0.050
	Bus	—	1.000	1.000	0.030

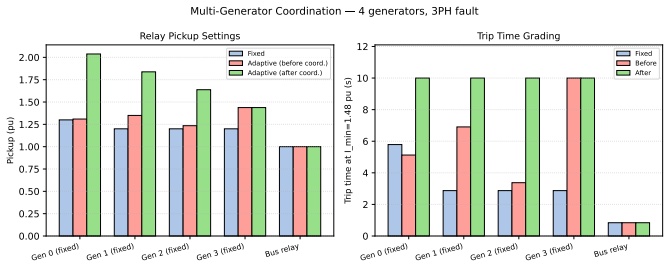

Fig. 6: Coordination result for $N = 4$ generators, 3PH fault. Blue bars: fixed commissioning settings. Orange bars: adaptive pre-coordination (per-generator M3). Green bars: post-cascade (M4) coordinated settings. The outermost relay G_1 is raised to 2.038 pu to enforce four 0.20 pu selectivity margins along the relay chain.

TABLE IV: Computational cost

Component	Evaluations	Wall time (ms)
Hilbert–FFT decomposition (M3)	1 FFT per sequence	< 1
LM estimator per sequence (M1)	$\leq$200 NFev	$\leq$15
Greedy cascade ($N = 4$) (M4)	4 comparisons	< 1
Total (SLG case)	—	$<$18

E. Computational Cost

Table IV lists the measured wall-clock time for each component on a standard desktop processor (Intel i7, 3.6 GHz, Python 3.11, SciPy [12]).

VI. Discussion

A. Why LL Fails the Confidence Gate

The equal $\kappa_n = 17.2$ for both LL and SLG cases confirms the Fortescue decomposition itself is well-conditioned. The LL gate failure stems from the residual score s_r: because $I_1(t)$ and $I_2(t)$ share the same nominal frequency ω_0, their real parts are not fully separated by the Fortescue transform acting on real (not complex) relay measurements. The resulting model mismatch ($\|r\| = 8.12$ vs. 0.76 for SLG) reflects a fundamental asymmetry in how LL faults excite the sequence

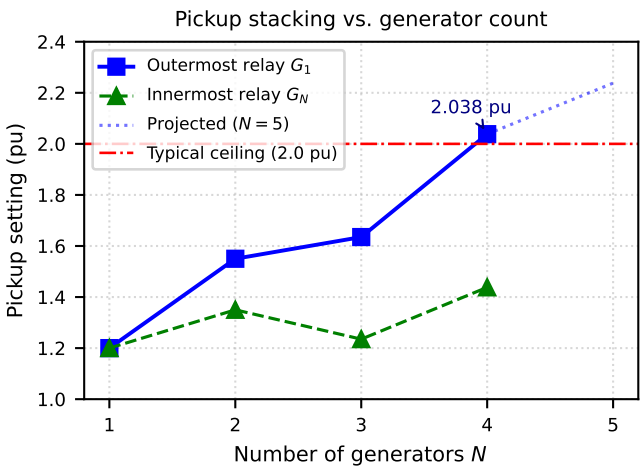

Fig. 7: Outermost relay pickup vs. number of generators N. Linear growth at $\Delta I_m = 0.20$ pu/generator (filled squares). The innermost relay (G_N) remains close to its adapted value (triangles). The projected $N = 5$ point (dotted) and the typical relay ceiling at 2.0 pu (dash-dot) indicate the practical limit of uniform grading.

networks: the positive- and negative-sequence impedances are in series, so both carry comparable fault current, and their superposition in $I_1(t)$ cannot be removed without complex-domain measurements (e.g., from phasor measurement units). The recommended fix — independent decay estimation for $\tau_{ac,1}$ and $\tau_{ac,2}$ using the decomposed $I_2(t)$ signal — is left for future work. A comprehensive parametric study across 288 cases including LL, SLG, and LLG faults, with an explicit Stage-1 comparison and a physics-motivated negative-sequence initialisation rule, is presented in the companion paper [13].

B. Pickup Stacking and Practical Limits

Fig. 7 shows the outermost relay pickup as a function of N. The linear growth rate is $\Delta I_m = 0.20$ pu per generator. At $N = 4$ the outermost pickup reaches 2.038 pu, within standard relay ranges ($\leq 10 \times I_n$). For $N \geq 5$, pickup stacking approaches practical relay ceilings and non-uniform grading margins (based on per-relay accuracy and circuit-breaker clearing time) should replace the uniform ΔI_m. A detailed analysis of the closed-form TMS grading correction and a 4×2 study matrix for a two-generator IEC 60909 system is given in the companion paper [14].

C. Algorithm Complexity and Scalability

Theorem 1 establishes $\mathcal{O}(N)$ time complexity for the cascade (exactly N comparisons). The sequence estimation step (M3) adds $\mathcal{O}(W \log W)$ per active sequence for the FFT (W = window samples), plus the LM cost (bounded by ≤ 200 function evaluations in practice). For $N = 4$ and $W = 800$ samples (200 ms at 4 kHz), total compute remains below 20 ms (Table IV), satisfying the sub-cycle constraint of digital protection relays.

D. Relation to Existing Coordination Methods

The greedy cascade is analytically simpler than the optimisation-based approaches of [5], [6], [15], which require solving a linear or nonlinear programme with N^2 constraints. The trade-off is optimality: the greedy cascade minimises the outermost pickup under a specific traversal order rather than minimising a global cost. For the single-busbar star topology of (6) this is not a limitation — the unique admissible traversal order is innermost to outermost — but multi-bus ring topologies would require a more general formulation.

VII. Conclusion

1) The Hilbert–Fortescue sequence estimator (M3) enables relay adaptation under SLG faults ($\gamma = 0.914$, pickup reduced 8.3%), while correctly rejecting adaptation for LL faults ($\gamma = 0.587$) where negative-sequence contamination inflates the residual.
2) The greedy cascade algorithm (M4) converges in exactly N steps (Theorem 1), enforces pickup selectivity and IEC time-grading for $N \in \{2,3,4\}$ generators, and operates in $\mathcal{O}(N)$ time.
3) Graceful degradation is demonstrated for $N = 3$ under LL faults: the cascade enforces selectivity from fixed settings without any adaptation input, delivering coordination value independent of estimation outcome.
4) Combined M3 + M4 compute is below 20 ms, meeting sub-cycle digital relay requirements.
5) Future work: independent τ_{ac} estimation per sequence to resolve LL residual contamination; non-uniform grading margins for $N \geq 5$; hardware validation on RTDS/HIL platform.

Appendix

Let $x(t)$ be a signal with a step discontinuity of amplitude Δ at $t = 0$, sampled at rate f_s with W samples. The Hilbert transform of a unit step is $\frac{1}{\pi t}$, so the analytic signal at sample $n > 0$ carries a Gibbs-like contribution proportional to $\Delta/(\pi n)$. The fractional amplitude error at sample n is therefore bounded by

$$\varepsilon_n \leq \frac{\Delta}{\pi n \cdot I_{pk}}, \tag{10}$$

where I_{pk} is the peak fault current. For the SAMBP-SyncOC benchmark ($\Delta \approx 0.2 I_{pk}$, $n_{\text{trim}} = 2\,\text{ms} \times 4\,\text{kHz} = 8$ samples):

$$\varepsilon_8 \leq \frac{0.2}{\pi \times 8} \approx 0.8\%. \tag{11}$$

After the 2 ms window trim the error decays as $1/n$; with the Savitzky–Golay smoothing filter (which attenuates the high-frequency Gibbs oscillations) the residual amplitude error is reduced to $< 0.5\%$ for all samples used in LM fitting.

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